Module 9, Lesson 1

AI 101 FOR MECHANICAL ENGINEERS:

Lines, neighbours and separating boundaries

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Lesson Structure

1. Parametric Models
2. Geometric Boundaries
3. Application Constraints

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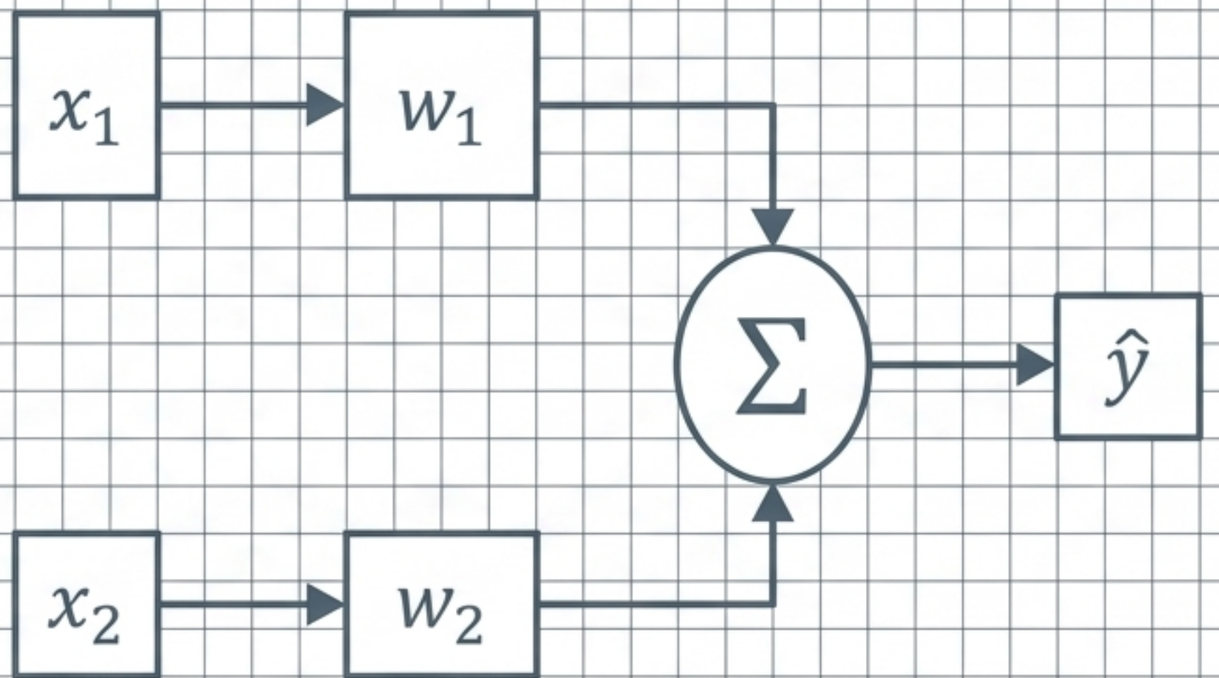
1. Parametric Models

Purpose: Define weighted combinations and probability-based linked predictive techniques.

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Linear Regression

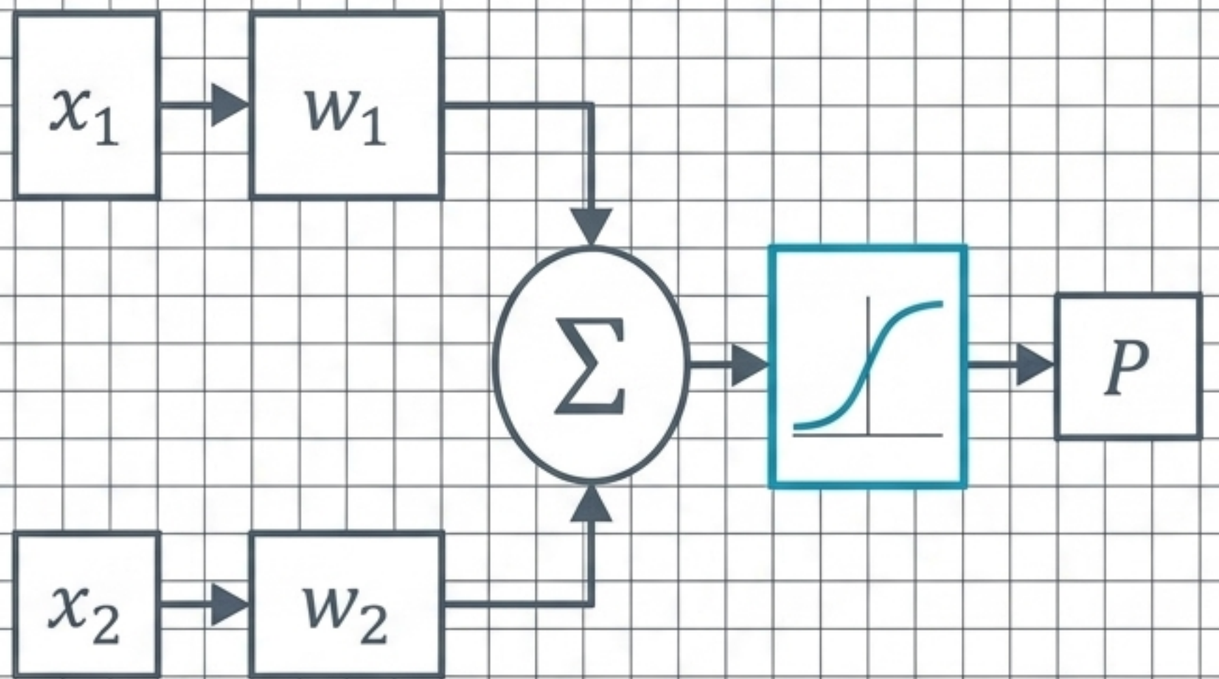
- Predicts a numerical value using a weighted combination.
- Model name alone does **not establish baseline performance.**



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Logistic Regression

- Commonly used to execute binary classification tasks.
- Estimates class probabilities utilizing a mathematical link function.
- **Example**: Estimating the probability of a binary fault.



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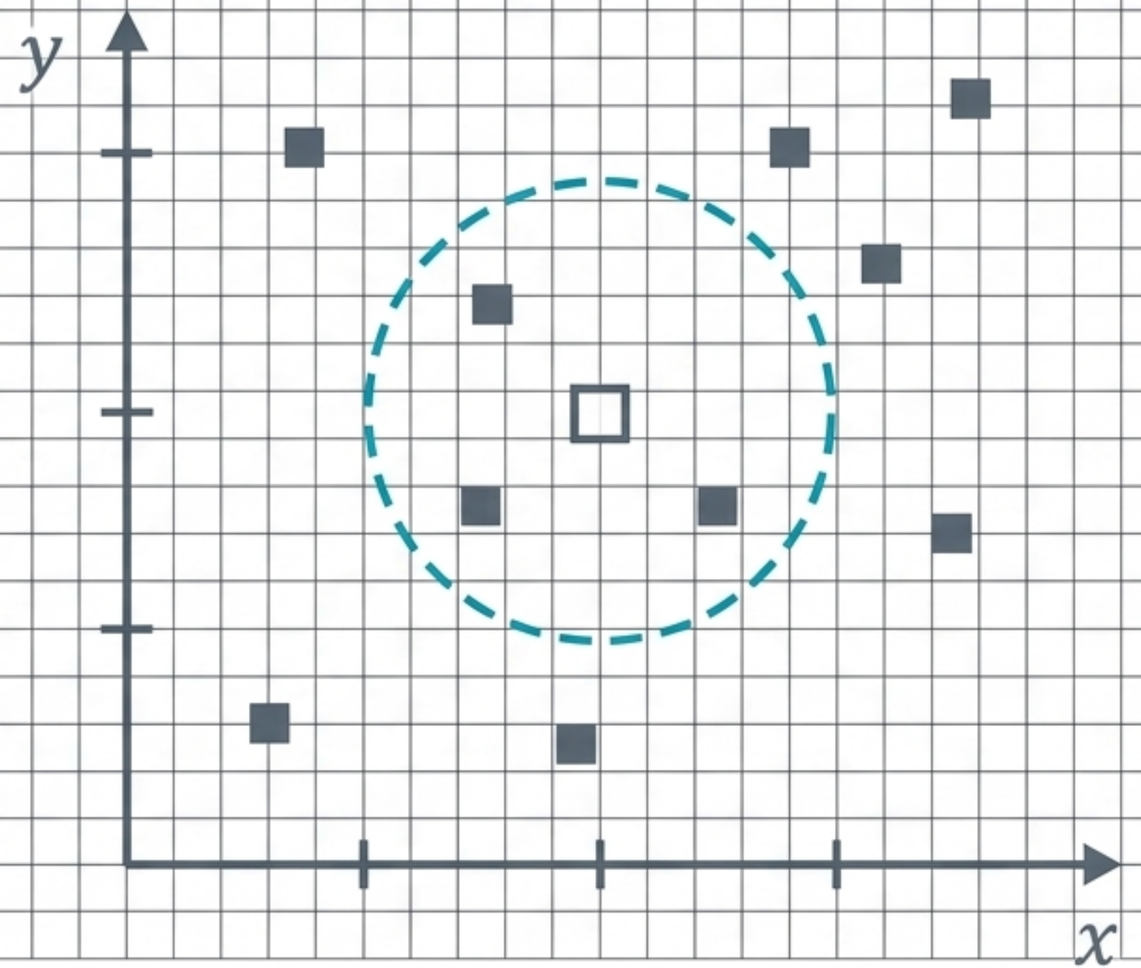
2. Geometric Boundaries

Purpose: Examine distance-based proximity and margin-based spatial separation.

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k-Nearest Neighbours (kNN)

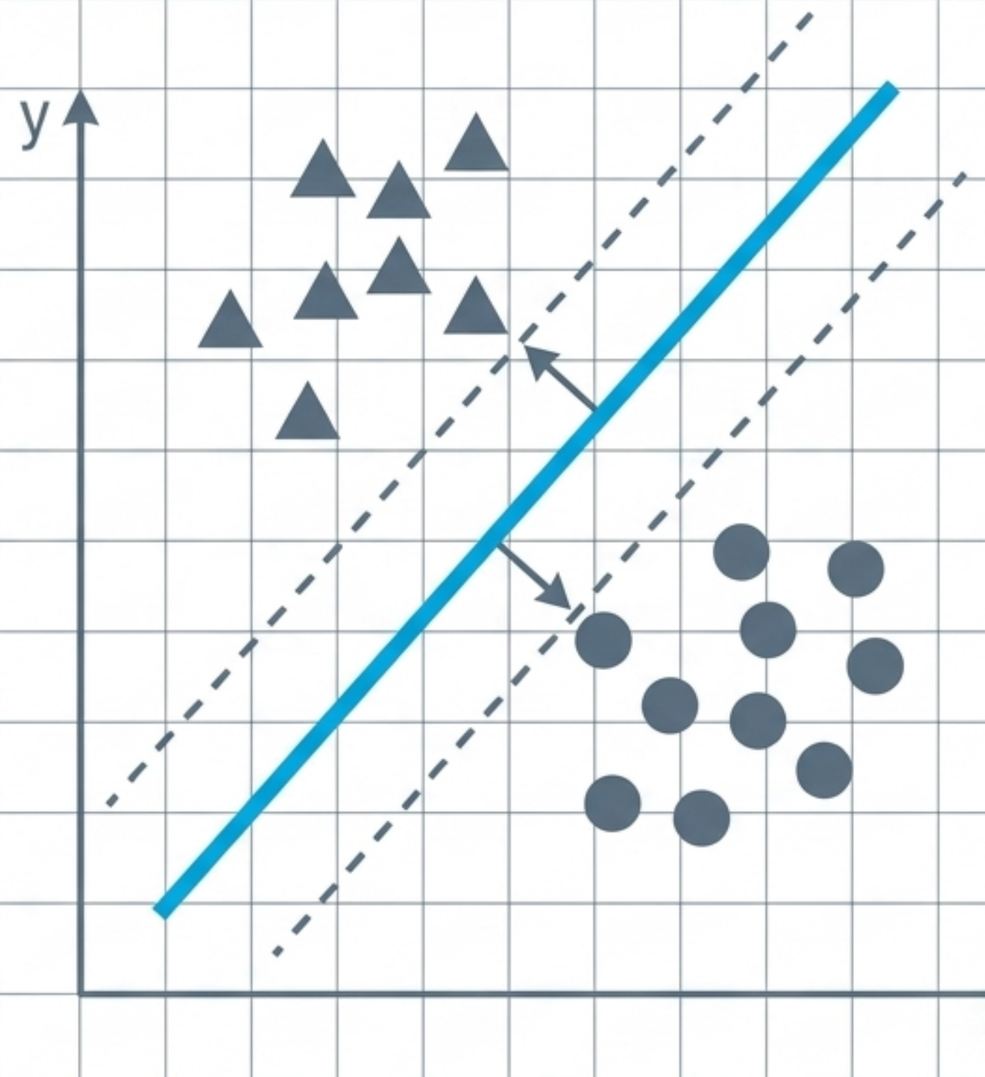
- Estimates an output using spatially nearby data examples.
- Execution relies entirely on a defined distance rule.
- **Example**: Estimating power generation from similar operating points.



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Support Vector Machines (SVM)

- Learns a defined separating boundary for data classification.
- Utilizes a geometric margin between differing data classes.
- Kernels allow the method to express nonlinear relations.
- Related support-vector techniques can also perform numerical regression.

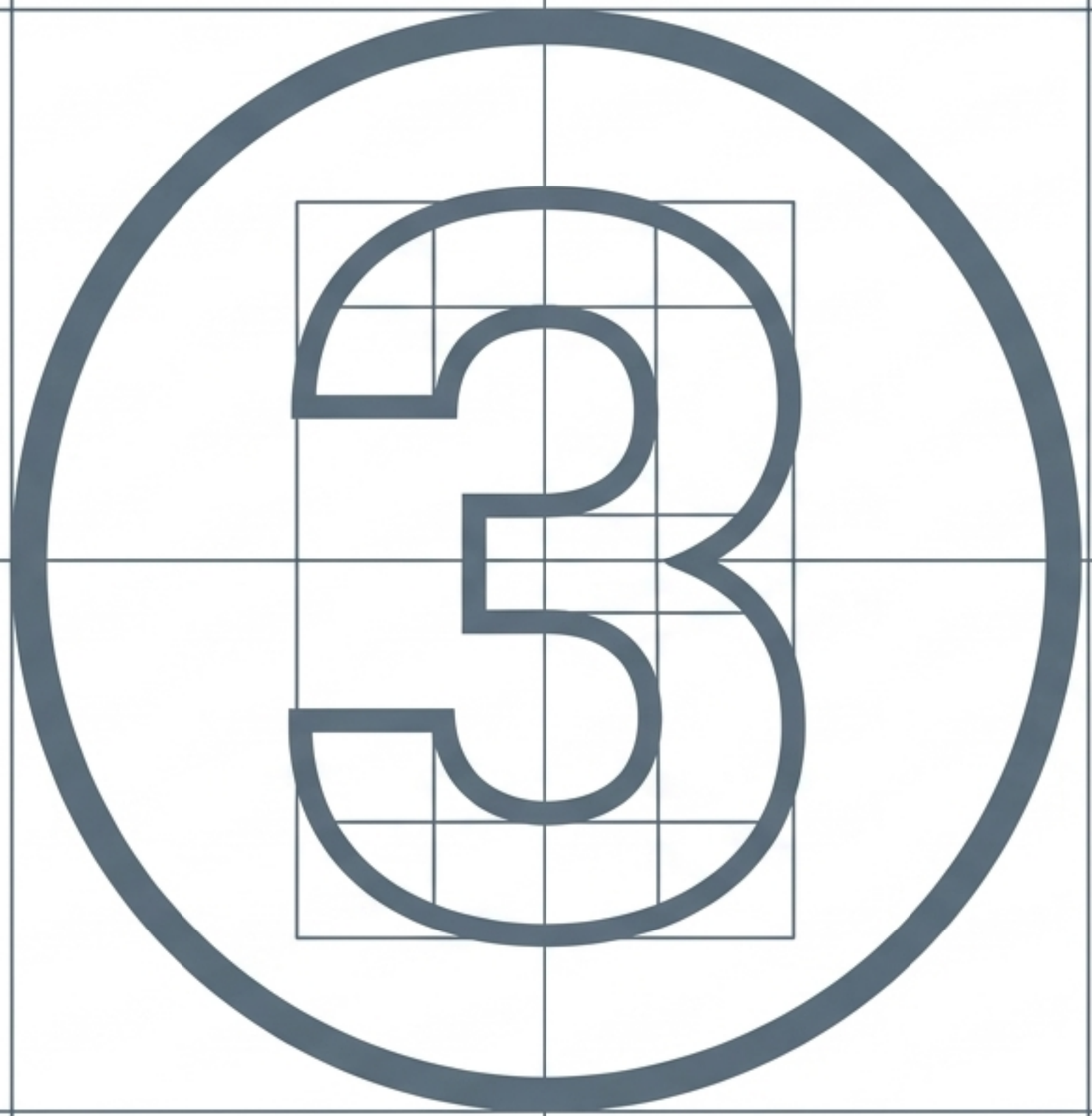


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Predictive Method Diagnostic Matrix

Method	Typical Purpose	Representation	Engineering Example
Linear Regression	Numerical prediction	Weighted combination	Continuous pressure
Logistic Regression	Classification	Probability through a link	Binary fault probability
kNN	Nearby-example prediction	Distance in feature space	Power from operating points
SVM	Separating boundary	Margin and optional kernel	Margin-based boundary

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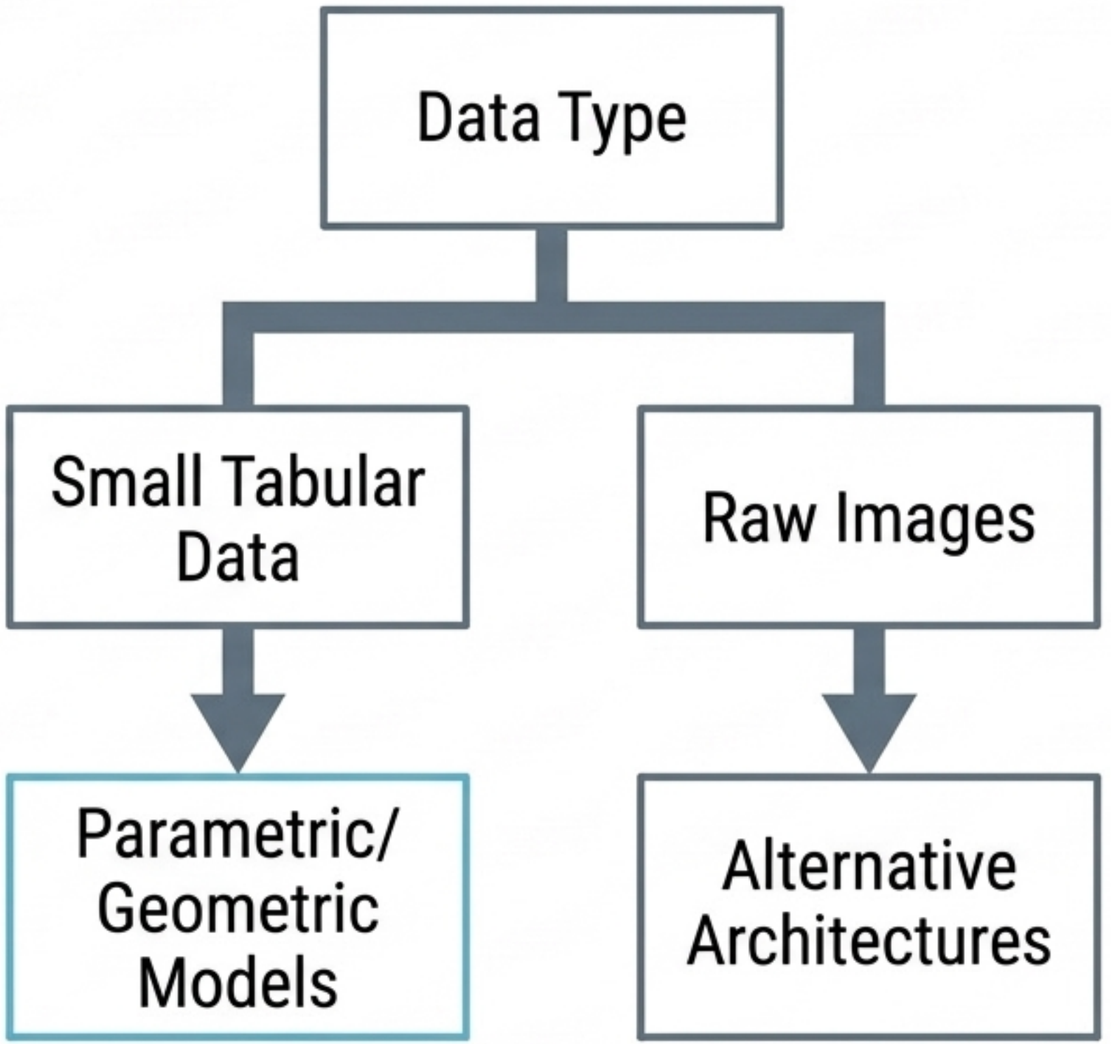


3. Application Constraints

Purpose: Assess data selection criteria and unit-scaling mechanics.

Data Structure and Method Selection

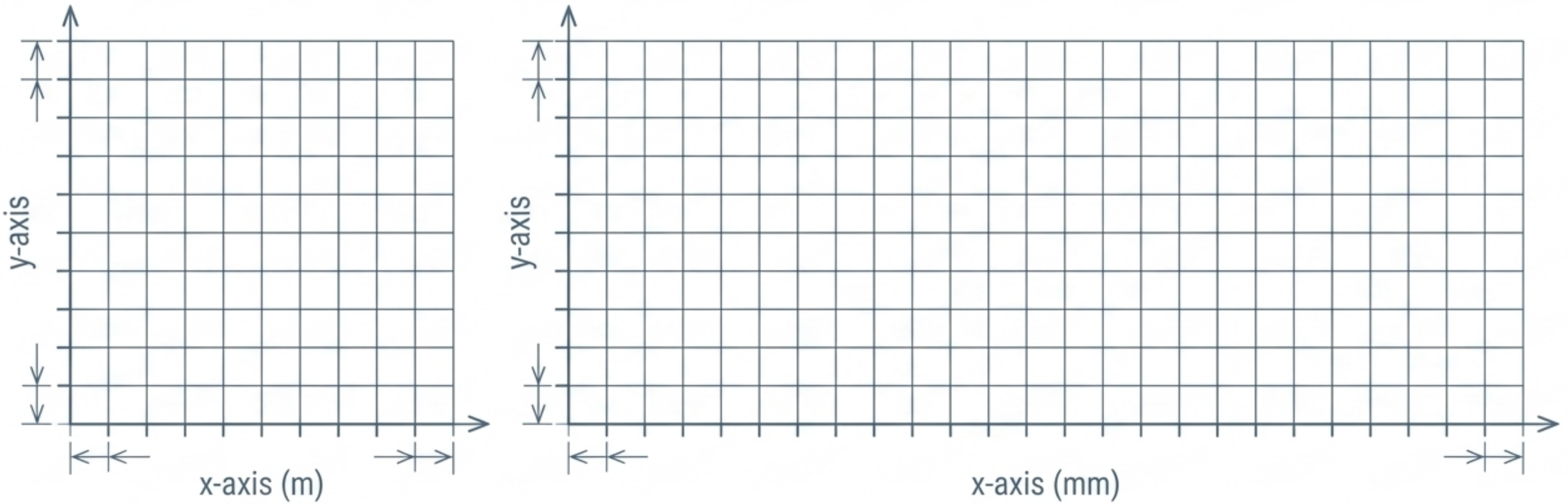
- Selection depends on the specific required output format.
- Consider the assumed data structure and total volume.
- Define how unseen performance will be quantitatively checked.
- Small tabular datasets require different methods than images.



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The Mechanics of Unit Scaling

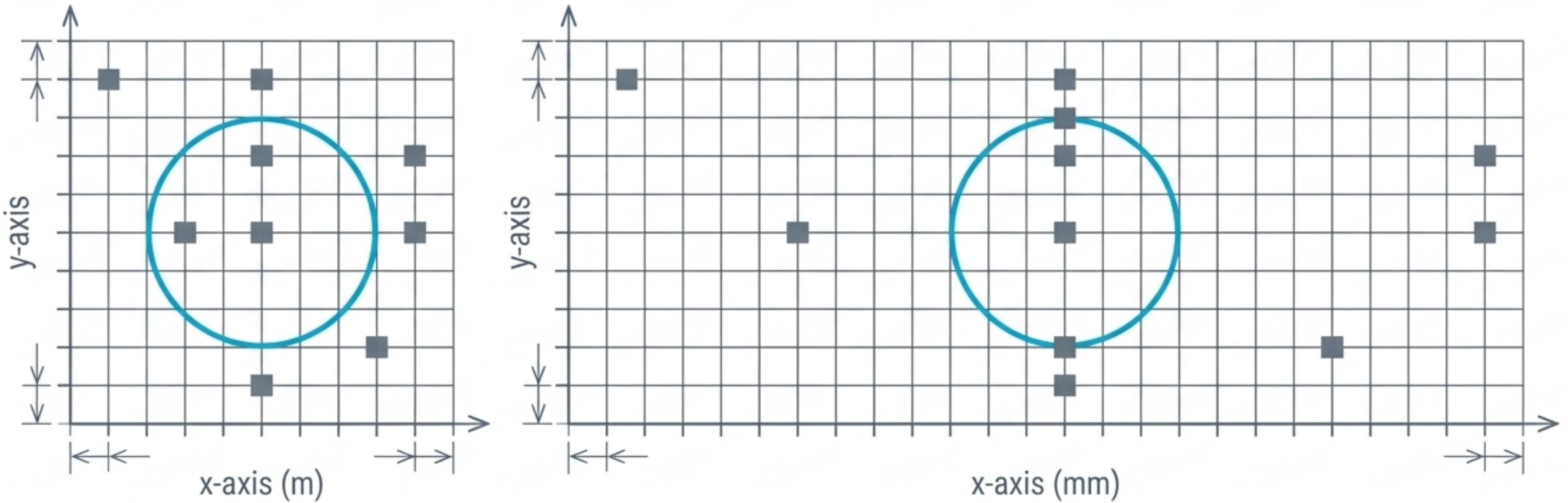
- Physical length features can switch from m to mm.
- The underlying physical lengths remain exactly the same.
- The numerical distance value multiplies by a factor of 1,000.



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Scaling Impact on Predictive Boundaries

- Without appropriate scaling, numerical distances distort feature spaces.
- Nearest neighbour rankings change due to coordinate stretching.
- Distance-based comparisons mandate strict dimensional unit standardisation.



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Key Points

1. Linear and logistic regression predict outputs via weighted combinations and link functions.

2. kNN and SVM classify data using spatial proximity and margin-based separating boundaries.

3. Distance-based algorithmic models are highly sensitive to physical unit scaling.

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